

AI Professor Assistant using WhatsApp, AI Agents and Workflow Automation

Project Overview

This project proposes an AI-powered virtual assistant that acts as an intelligent bridge between students and professors. Instead of contacting professors directly, students communicate with the AI assistant through WhatsApp. The assistant understands the student's request, checks the professor's availability, schedules meetings, and sends notifications automatically.

Problem Statement

Professors are often busy with classes, research, meetings, and administrative work. Students spend time waiting for replies or repeatedly sending messages. The proposed system automates this communication while allowing professors to approve or reject appointments.

Objectives

- Automate appointment scheduling
- Reduce manual communication
- Provide 24/7 AI support
- Improve response time
- Maintain appointment history in MongoDB

Technology Stack

WhatsApp Cloud API, n8n, AI Agent (OpenAI/Llama), FastAPI, MongoDB, Google Calendar API, Docker.

System Workflow

Student → WhatsApp → WhatsApp Cloud API → n8n Workflow → AI Agent → MongoDB & Google Calendar → Professor Approval → Student Notification

AI Agent Responsibilities

Understand requests, detect meeting intent, ask follow-up questions, prioritize urgent requests, suggest available time slots, maintain conversation history, and answer common FAQs.

Automation Features

n8n automatically checks availability, updates Google Calendar, stores appointment details in MongoDB, sends WhatsApp and email notifications, and reminds both the student and professor before meetings.

Future Enhancements

Voice assistant, multilingual support, ERP integration, analytics dashboard, and AI-generated meeting summaries.

Conclusion

This project combines Generative AI, workflow automation, APIs, and cloud technologies to build an intelligent academic assistant that saves time for both students and professors while improving communication efficiency.